

Patuakhali Science and Technology University
Department of Computer Science and Information Technology
B.Sc. Engg.(CSE) Level-4 Semester II Final Examination of Session 2018-19, July-December-2022
Course Code: CIT 421 Course Title: Computer Graphics and Image Processing
Credit Hour: 3.0 Full Marks: 70 Duration: 3 Hours.

[Figures in the right margin indicate full marks. Split answering of any question is not recommended. Write the full question number e.g. 1(B) (i) before the answer paragraph]
[Answer any 5 of the following questions]

- 1 (a) Define display processor. Which purposes does the storage tube serves? 03
Describe the triad arrangement of red, green, and blue guns in color CRT. 02
- (b) What is scan conversion? Write down the advantage of developing algorithms for scan conversion. 02
- (c) Describe the process of scan conversion of a straight line. Write down the properties of a good line drawing algorithm. 02
- (d) If a line is drawn from (2, 3) to (6, 15) with use of DDA, how many points will be needed to generate such line? 02
- (e) Derive the process of drawing a line using Bresenham's Line drawing algorithm with exact mathematical background. 05
- 2 (a) Define a circle using polar co-ordinates. Write down the algorithm for drawing a circle using polar co-ordinates. 03
- (b) Derive the process of drawing a circle using Bresenham's circle drawing algorithm with exact mathematical background. 05
- (c) Write down the drawback of polynomial method for scan conversion of an ellipse. Describe the trigonometric method for scan conversion of an ellipse. 03
- (d) Write down the midpoint ellipse algorithm. 03
- 3 (a) What is composite transformation? Describe the procedure of general pivot point rotation or rotation about fixed point. 03
- (b) Define homogeneous coordinates. Write down the matrices for two-dimensional transformation in homogeneous coordinate. 04
- (c) Define clipping. Write down the point clipping procedure. 02
- (d) Describe in detail the Cohen-Sutherland algorithm to clip a straight-line segment against a rectangle. 05
- 4 (a) i) Illustrate and describe the Sampling and Quantization with example. 06
ii) The pixel values of the following (2x2) image are represented by 8-bit integers. Determine I with a gray_level resolution of 2^k for $k=3$.

I =

123	162
137	15

- (b) i) Why do we need to detect edge in images? What method is used by Homogeneity operator and Difference operator for detecting edges? 04

ii) Apply Homogeneity operator and Difference operator in the following image.

12	13	15
16	18	20
22	21	19

- (c) What is Image Histogram? What information the shape of the histogram gives us? 04

- 5 (a) i). Find the histogram of the following (6x6) sub image and then find Min, Max gray level values as well as minimum and maximum pixel values in this image. 06

12	12	12	12	12	12
8	16	16	16	16	22
8	16	10	17	16	22
8	16	17	10	16	22
8	16	16	16	16	22
20	20	20	20	20	20

ii) Apply the histogram equalization over the above (6x6) sub image.

- (b) Why color is important in image? List and define Color distinguishing characteristics. 04
- (c) Describe additive and subtractive color models. 04
- 6 (a) What is noise in image? Describe Gaussian and Salt & Pepper noise. 04
- (b) Why filtering is used in image processing? What are the effects of the Coefficients of the convolution masks in spatial filters? 04
- (c) Define maximum and minimum filters. Use a (3x3) Mean filter for removing noise from the following sub image. 06

I =

8	9	10	11	12
12	8	9	10	11
13	12	8	9	10
14	13	12	8	9
15	14	13	12	8

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Patuakhali Science & Technology University (PSTU)	
Department of Computer Science and Information Technology (CSIT)	
Final Examination: July-December 2022	
Course Code: CIT 422	Course Title: Computer Graphics and Image Processing Sessional
Session: 2018-19, Program: B.Sc. Engg.(CSE), Semester: 8 th	
Marks – 70	

[Answer the marked questions/]

Section A		
1.	Write a program to draw line using Bresenham's algorithm.	20
2.	Write a program to draw Circle using Bresenham's algorithm.	20
3.	Write a program to Draw a line using DDA (Digital Differential Analyzer) algorithm.	20
4.	Write a program to draw Ellipse using Bresenham's algorithm.	20
5.	Write a program to apply line drawing with simple line equation	20
6.	Write a program to apply circle drawing with polar coordinates.	20
7.	Write a program to apply rotation and reflection to 2-Dimensional shapes.	20
8.	Write a program to clip the lines fallen outside the window using cohen Sutherland line clipping.	20
Section B		
1.	Write a program from scratch to find the Edges of an image using Sobel Operator.	20
2.	Write a program from scratch to remove noise from an image using Mean Filter.	20
3.	Write a program from scratch to remove noise from an image using Median Filter.	20
4.	Write a program from scratch to remove noise from an image using Mid-point Filter.	20
Section C		
1.	OpenGL Graphic Project	20
2.	Viva-Voce	10

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Patuakhali Science and Technology University

B.Sc. Engg. (CSE) Level-4, Semester-II, Final Examination July-December 2022

Course code: CIT-423

Course Title: Machine Learning

Session: 2018-2019

Credit hour: 3.00

Full marks: 70

Duration: 3 hours

[Figures in the right margin indicate full marks. Split answering of any question is not recommended]

Answer any 5 from the following questions.

1. (a) Why is dimensionality reduction important in machine learning? Give the different methods of dimensionality reduction. 2+2
- (b) i. Explain features selection and features extraction with appropriate example. 3+2
 ii. Shortly explain the major aspects related to the curse of dimensionality. 5
- (c) Find out principal component from the following set.

	Feature 1	Feature 2	Feature 3
Sample 1	3	3	2
Sample 2	4	5	3
Sample 3	3	1	4

2. (a) Distinguish between PCA and factor analysis. Explain the domains of curse of dimensionality. 2+2
- (b) In computational system, why are metaheuristics approaches important? – Explain it. Also give the challenges to deploy the metaheuristics approaches. 3+2
- (c) i. What is optimization? Give two real examples of optimization problem. 3+2
 ii. How does the GA help to solve optimization problem?

3. (a) i. Define reinforcement learning. Mention its limitations. 2.5+2.5
 ii. Explain the elements of Reinforcement Learning. 2
- (b) Distinguish among supervised, unsupervised and reinforcement learning. 3
- (c) i. Shortly mention various swarm intelligence with features which are currently employed in computation. 3+3
 ii. Explain- How could we implement any swarm intelligence into computational problem solution?

4. (a) Define the following terminologies: hypothesis class in classification, problem of generalization, most specific hypothesis, overfitting, and cross-validation. 4
- (b) What is the VC dimension? Illustrate different situations of VC dimensions. Show regression as a supervised learning problem with an example. How to estimate the parameters (w_1, w_0) of regression by using the error (loss) function? 6
- (c) What is mixture density? Explain the component densities as an unsupervised learning method. How does the k-means algorithm work? Show that the k-means algorithm is highly sensitive to the initial seed. 5

5. (a) Write the advantage of the parametric model. Mention the function of maximum likelihood estimation in the parametric model. Briefly explain Gaussian distribution as a parametric model. 5

- State bias and variance for evaluating an estimator. Derive the mean square error with bias and variance. 4
- (c) What is parametric classification? Describe the parameter estimation process of parametric classification. How to assign an instance, x , to the class with the nearest mean in the discriminant function. 5
- 6 (a) Write the scope of artificial neural networks for future research trends in deep learning. 2
- (b) Distinguish the perceptron training rule and Delta rule in learning neural networks. State the stochastic gradient descent version of the backpropagation algorithm for feedforward networks containing two layers of sigmoid units. 5
- (c) What is convolutional neural network, CNN? Write the convolution operation of a two-dimensional image and a two-dimensional kernel. What is the problem of long-term dependency? How to solve the long-term dependency problem using long short-term memory, LSTM, neural network. 7

Dept. of Computer and Communication Engineering

Faculty of Computer Science and Engineering

Patuakhali Science and Technology University

Dumki, Patuakhali-8602, Bangladesh

Final Examination of B. Sc. Engineering in CSE Level: 4 Semester: II Session: 2018-2019

Course Code

Course Title

July-December 2022

Credit: 03

CCE 423

Wireless and Cellular Communication

Time: 03 Hr

Answer any 05 out of 06 Questions (Split answers are highly discouraged)

Marks: 70

- 1 [A.] Provide a historical overview of the evolution of wireless and cellular communication technologies in Bangladesh in a tabular format. 4
- [B.] Currently, network operators roll-out 5G technology. Figure out the strategies of the different network operators in Bangladesh while migrating towards 5G systems. Can you find reasons why a single common system is not in sight? 3
- [C.] What are the main differences between GSM and ISDN at a high level, considering aspects such as technology and purpose, communication type, network structure, service integration, and usage context? 4
- [D.] What are the different characteristics that a communication device can exhibit, and how do these characteristics impact its functionality and use? 3
- 2 [A.] i) Name some key features of the GSM, DECT, TETRA, and UMTS systems. 6
ii) Which features do the systems have in common?
iii) Why have the three older different systems been specified?
iv) In what scenarios could one system replace another?
v) What are the specific advantages of each system?
- [B.] What are the key mechanisms and algorithms used for data encryption in GSM, and how do they contribute to ensuring the security and privacy of communications in a GSM network? 4
- [C.] Why are so many different identifiers/addresses (e.g., MSISDN, TMSI, IMSI, PIN) needed in GSM? Give reasons and distinguish between user-related and system related identifiers. 2
- [D.] Draw the flowchart of Mobile terminated call (MTC). 2
- 3 [A.] In a cellular system with a total of SS duplex channels that are divided among NN cells, with each cell receiving a unique and disjoint group of kk channels, how is the total number of available radio channels expressed? Discuss the impact of cluster size NN on co-channel interference and the frequency reuse distance. Additionally, explain how to determine the nearest co-channel neighbors in a hexagonal cell layout with an example. 4
- [B.] If a total of 33 MHz of bandwidth is allocated to a particular FDD cellular telephone system which uses two 25 kHz simplex channels to provide full duplex voice and control channels, compute the number of channels available per cell if a system uses (a) 4-cell reuse, (b) 7-cell reuse (c) 12-cell reuse. If 1 MHz of the allocated spectrum is dedicated to control channels, determine an equitable distribution of control channels and voice channels in each cell for each of the three systems. 4
- [C.] i. What are some of the key performance indicators for 6G networks, and what technologies will enable these targets? 4
ii. What are the key challenges associated with the deployment of mmWave technologies in 5G and 6G networks?
- [D.] How is the spectrum allocated for 5G deployment, and what are the characteristics of each frequency band used in this allocation? 2

- 4 [A.] i. Name several methods for ISI mitigation. 4
 ii. How does ISI depend on the carrier frequency, symbol rate, and movement of sender/receiver?
 iii. What are the influences of ISI on TDM schemes?
 [B.] Why mobile network providers install several thousands of base stations throughout a country (which is quite expensive) and do not use powerful transmitters with huge cells like, e.g., radio stations? 4
 [C.] Considering duplex channels, what are alternatives for implementation in wireless networks? 3
 What about typical wired networks? 3
 [D.] Define data carousel? Give a short explanation on different types of data carousel.
 5 [A.] Think of a phase diagram and the points representing bit patterns for a PSK scheme (see figure 1). 4

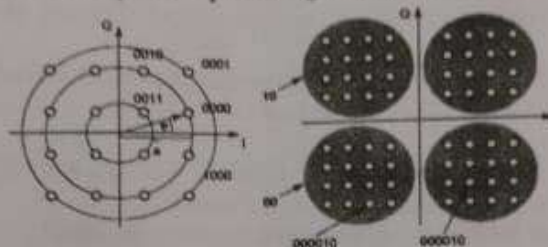


Figure 1

- i. How can a receiver decide which bit pattern was originally sent when a received 'point' lies somewhere in between other points in the diagram?
 ii. Why is it difficult to code more and more bits per phase shift?
 [B.] Explain the term interference in the space, time, frequency, and code domain. What are countermeasures in SDMA, TDMA, FDMA, and CDMA systems? 4
 [C.] How could location based services and broadcast systems work together? 2
 [D.] Illustrate the radiation patterns of isotropic radiator, simple dipole, directed antenna and sectorized antennas with sector 6. 2
 [E.] How can MACA still fail in case of hidden/exposed terminals? Think of mobile stations and changing transmission characteristics. 2
 6 [A.] i. What are the advantages of a fixed TDM pattern compared to random, demand driven TDM? 6
 ii. Compare the efficiency in the case of several connections with fixed data rates or in the case of varying data rates.
 iii. Now explain why traditional mobile phone systems use fixed patterns, while computer networks generally use random patterns.
 iv. In the future, the main data being transmitted will be computer-generated data. How will this fact change mobile phone systems?
 [B.] How the MOT protocol solve the problem faced for broad range of different receiver capabilities? 3
 [C.] Why can waves with a very low frequency follow the earth's surface? Why are they not used for data transmission in computer networks? 3
 [D.] Illustrate the components of a DAB sender. 2

Patuakhali Science and Technology University

Department of Computer Science and Information Technology (CSIT)

8th Semester (Level-4, Semester-II), Mid-Term Examination of B.Sc. Engg. (CSE)

Course Code: CIT-421 Course Title: Computer Graphics and Image Processing

Credit Hour: 3.0 Full Marks: 15 Session: 2018-2019 Time: 01 Hour

$$x = n + \frac{1}{m}$$

[Figures in the right margin indicate full marks. Write answer to the following questions]

1. What is digital image? Why do we need to process image? 02

2. What is Edge in digital image? What are the two basic principles for an edge detector mask? 02

3. What is Sobel edge operator? Calculate the edges magnitude matrix of the following sub image using Sobel edge detector. 03

Row Mask			Column Mask		
-1	-2	-1	-1	0	1
0	0	0	-2	0	2
1	2	1	-1	0	1

1	2	2	1
1	1	0	3
2	4	1	5
2	1	2	0

4. Explain with diagram raster scan display technique. Differentiate between Vector scan display and Raster scan display. 2

5. Explain the principles of operation of each of the following.
(ii) Liquid crystal displays (ii) Cathode Ray Tube displays 3

[Note: You may illustrate your answers with a diagram.]
6. Consider the line from (0, 0) to (4, 6). Use DDA algorithm to rasterize this line. 3

$$\frac{6}{4}$$

$$A = \frac{6}{4}$$

ALCD CRT
positive

visualize

Patuakhali Science and Technology University
Department of Computer Science and Information Technology

8th Semester (Level-4, Semester-II), Midterm Examination of B.Sc. Engg.(CSE), July-Dec./2022, Session: 2018-19

Course Code: CIT-423 Course Title: Machine Learning

Full Marks: 15 Duration: 1 hour

1. Write a formal definition of the classification problem. Explain the classification problem from credit scoring data with low-risk and high-risk customers. Derive a classification rule for the hypothesis class from the family car example. 04
2. What is market basket analysis? How to determine the purchased trend using association rule? Formulate a Bayesian classification model where Bayes' classifier chooses the class with the highest posterior probability. 04
3. Define dimensionality and way of dimensionality reduction. Why is dimensionality reduction important? 03
4. Find out principal component from the following set. 04

	Feature 1	Feature 2	Feature 3
Sample 1	5	4	2
Sample 2	3	6	8
Sample 3	9	2	4

Patuakhali Science and Technology University

8th Semester (Level-4, Semester-II), Mid Examination of B.Sc. Engg. (CSE), July-December: 2022

Course Code: CCE-423 Course Title: Wireless and Cellular Communication

Credit Hour: 3.0 Full Marks: 15 Session: 2018-2019

- 1 Explain the key differences between wireless communications and cellular communications, highlighting how each technology is utilized in modern communication systems. 3
- 2 i) "Are we currently operating at the Shannon limit in the physical layer of communication systems? What would Shannon's perspective be on this? 4
ii) Additionally, what factors contribute to the reduction in Wi-Fi performance?" How can we overcome this limitation?
- 3 What are the means to mitigate narrowband interference? What is the complexity of the different solutions? 2
- 4 Spreading the spectrum can be achieved in two different ways. Differentiate between them. 3
- 5 What are the main reasons for using cellular systems? How is SDM typically realized and combined with FDM? How does DCA influence the frequencies available in other cells? 3

DSSS

TDM + CDM

FDM + TDM

FHSS

FDMA

Hopping

Sequence

use hopping

FAIR